List of revisions made to "On Vassiliev Invariants and Weight Systems of Classical and Welded Knots"

There are two reports for this thesis, a report by Audoux and a report by an anonymous reviewer.

The report by Audoux contains issues to be corrected / suggestions, of which six are identified as major, and twenty-nine are identified as minor. The report also lists some possible typos for correction. We will reference these issues as Audoux.M1 through Audoux.M6 for the major issues, and Audoux.m1 through Audoux.m29 for the minor issues.

The report by Raynor contains two general required revisions in Section 2.1, a list of required revisions by section of the thesis, and a list of recommendations for future publication of the thesis, also by subsection of the thesis. We refer to the revisions in this report by the sections of the report, then by the sections of the thesis which they address.

Table 1: List of revisions made based on corrections, suggestions or comments in review reports

Position in submitted thesis	**Position in revised thesis**	**Modification made**	**Relevant issue(s) in report(s)**
Paragraph after Remark 1.1.2	Paragraph after Remark 1.1.2	Be explicit about use of phrases "positive and negative crossings" when only dealing with embeddings in $\mathbb{R}^3$.	Audoux.m1
-	Definition 1.1.3	Define knot invariant and provide reference for definition of ambient isotopy. Assert that everything in the text is done over $\mathbb{Q}$.	Audoux.M1
Start of Section 2.2	Start of Section 2.2	Small changes to wording to make meaning clearer	Audoux.M2
-	Definition 2.2.1	Give 'tensor' its own environment-level definition	Audoux.M2
-	Proposition 2.2.2 and its proof	Provide details for the isomorphism between multilinear maps and elements of the space $X_1^* \otimes \cdots \otimes X_n^* \otimes Y_1 \otimes \cdots \otimes Y_m$	Audoux.M2
First diagram of Section 2.2, diagrams of section 3.3	First diagram of Section 2.2, diagrams of section 3.3	Remove dual notation (e.g. x^*, y^*) from string diagrams, as the top-to-bottom convention instead suffices to determine whether they are inputs or outputs	Audoux.M2
Sections 2.2, 2.3	Sections 2.2, 2.3	Remove arrows from string diagrams where they're unnecessary, adressing confusion	Audoux.M2
-	Definition 2.2.4	Definition environment for metric Lie algebra	Audoux.M2
Question 2.2.7 Question 1.5.12	Remark 2.2.11 Remark 1.5.12	Change environment type of open questions from **Question** to **Remark**. This addresses the point in Raynor.2.1 about the use of the Question environment, as now (using the submitted version numbering) 1.2.2 and 2.2.9 describe natural questions to explore and pre-empt what follows, and remain as Question environments. Open problems which are given as curiousities and are not further explored in the thesis take the form of Remark environments	Raynor.2.1

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Table 1: List of revisions made based on corrections, suggestions or comments in review reports (Continued)

Third last paragraph of Section 2.3	Third last paragraph of Section 2.3	Emphasise that the theory of Section 2.3 leaves Question 2.2.9 (submitted numbering) an open question. The questions still explores and pre-empts Section 2.3, so remains as a Question environment.	Raynor.2.1
Paragraphs preceeding and proceeding Definition 1.2.3	Paragraph preceeding and proceeding Definition 1.2.3	Emphasise that the ways in which Definition 1.2.3 will answer Questions 1.2.2 will be explained in the current section.	Audoux.m2
Defintion 1.2.4	Definition 1.2.4	Include a proper definition of the signs of a singularity, an example, and state its equivalence to defining a coorientation on the $(m+1)$st stratum	Audoux.m3 Raynor.1.2
Paragraphs preceeding Definition 1.2.5	Paragraphs preceeding Definition 1.2.5	Emphasise that the equation is true for any path	Audoux.m4
First paragraph of Section 1.3	First paragraph of Section 1.3	Remove unclear reference to linearisation/projectivisation.	Audoux.m6
First paragraph of Section 1.3	First paragraph of Section 1.3	Remove claim that Vassiliev invariants detect 3_1 implies that f_{3_1} is power series Vassiliev	Audoux.m7 Raynor.1.3
	Before Definition 1.4.8	Specify meaning of "finite-type"	Audoux.m8
Remark 1.5.5	Remark 1.5.5	Remove typo to address apparent self-referencial definition of universal Vassiliev invariant	Audoux.m9
-	Proposition 2.2.5 and proof	Make clear the proposition status, and prove the relation on a metric Casimir coming from an invarint metric	Audoux.m10
Example 2.2.3	Example 2.2.8	Insert explicit computation of the Casimir element	Audoux.m12
Section 2.3	Section 2.3, Remark 2.3.1	Move parts of the suggested paragraph into the section introduction as suggested. As a result, this becomes the main point of the introduction paragraph and the part about historical terminology is moved into a new remark.	Audoux.m13
Theorem 2.3.8 to end of subsection	Theorem 2.3.9 to end of subsection	Small rephrasings to increase clarity. However to my knowledge it is not in the literature whether Lieberum's order 15 example vanishes on Lie superalgebra weight systems.	Audoux m.14
Section 2.4	Section 2.4	State that $\mathfrak{g}_2$ and $\mathfrak{f}_4$ are the usual excpetional Lie algebras.	Audoux m.15
Example 2.5.3	Example 2.5.3	Fix typo, indeed vector space should be set	Audoux m.17
	Before Definition 2.5.4	Change 'there is no natural notion of tensor product for operads' to 'the tensor product is not a natural operation for operads', removing the ambiguous claim	Audoux m.18
Remark 2.5.6	Remark 2.5.6	Specify which functors are left- and right- adjoint	Audoux.m19
Definition 2.5.8	Definition 2.5.8	Add boldface text name "cyclic action condition" and rephrase to make obvious that the the relation is part of the definition of a cyclic operad	Audoux.m20

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Table 1: List of revisions made based on corrections, suggestions or comments in review reports (Continued)

Definition 2.5.9	Definition 2.5.9	Define the relevant symbols A and m in the same sentence	Audoux.m21
Paragraph before Definition 3.1.1 Definition 3.1.4	Paragraph before Definition 3.1.1 Definition 3.1.4	Accept the suggested wordings not to be factually misleading about welded knots	Audoux.m24
	Before Definition 3.1.3	Add suggested citation	Audoux.m25
Definition 3.2.5 Proposition 3.2.6 Section 3.2	Definition 3.2.5 - Section3.2	Define Jabobi diagrams as two-in-one-out by definition rather than proposition, remove all mention of the two-in-one-out rule	Audoux.M6, Audoux.m26
Section 3.3	Section 3.3	Clarify and reference citation that defines the semidirect product of $\mathfrak{g}$ on $\mathfrak{g}^*$ in the appropriate way	Audoux.m27
Definition 3.4.3	Definition 3.4.3	Specify as suggested that β is a generic bracket morphism, and so it has the same properties as α in Example 2.5.3	Audoux.m28
Entire text	Entire text	Fix typos	Both reviewers' identified typos except those identified in Table 2
Introduction paragraphs 6 and 8	Introduction paragraphs 6 and 8	List explicitly the new results of Chapters 2 and 3	Audoux.M3 Raynor.2.2
Example 2.5.11 Example 2.5.13 The surrounding paragraphs	Definition 2.5.11 Definition 2.5.13 The surrounding paragrahs	Change the examples to Definitions, as they are not yet defined.	Audoux.M4
-	Definition 3.2.2	Add definition environment for the boundary of a welded knot (and derivative of a welded knot invariant)	Audoux.M4
Construction 3.3.1	Construction 3.3.1	Remove reference to $I\mathfrak{g}$ which hasn't been defined yet	Audoux.M4
Lemma 2.5.20	Lemma 2.5.20	Make explicit that kS_n refers to the group algebra	Audoux.M4
After Definition 2.5.2	After Definition 2.5.2	Add paragraphs introducing the diagrammatics of operad operations and operad multiplication (concatenation)	Audoux.m16
-	Definition 2.5.3	Add definition environment for algebra over an operad including summary of the axioms, and reference to explicity written out axioms	Raynor.2.5 Audoux.m16
From the end of Definition 2.5.2 through to Example 3.5.3	From the end of Definition 2.5.2 through to Example 3.5.4	Remove inconsistencies about the role of the enriching category. Place explanatory material on the Lie operad outside the example environment.	Raynor.2.5 Audoux.m17
-	Proposition 2.5.5 and its proof	Add explicit statement that the algebras over the Lie operad are the Lie algebras	Raynor.2.5
Before Definition 2.5.9	Before Definition 2.5.11	Add the word object to better convey what the following definition is a generalisation of	Raynor.2.5

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Table 1: List of revisions made based on corrections, suggestions or comments in review reports (Continued)

Section 2.5	Section 2.5	Correct use of "inverse" and "invertible" to "adjoint" and "with adjoint"	Raynor 2.5
Theorem 2.5.19 and preceeding paragraph	Theorem 2.5.21 and preceeding paragraph	Make explicit why $\mathbf{1}_c$ is an algebra over the corresponding operad	Raynor 2.5
Definition 2.5.8	Definition 2.5.10	Make explicit the abuse of notation of writing simply σ_c for many σ_c's of different sizes, since which one is acting can always be read from context.	Audoux.m20
After Definition 2.5.8	After Definition 2.5.10	Add picture and explanation of the how the cyclic action condition relates to permuting inputs and outputs of the operations in an operad	Audoux.m20
Proof 2.5.19	Proof 2.5.21	Explain why every Jacobi diagram can be presented without local minima	Audoux.m22
Introduction to Section 2	Introduction to Section 2	Add introduction summarising the section's key points	Audoux.m23
End of Section 2.4	End of Section 2.4	Summarise the main result of Section 2.4	Audoux.m23
Definition 2.5.9	Definition 2.5.11	Fix the incorrect axiom S_{n+1}-invariant inherited from Hinnich-Vaintrob, replacing it with the correct invariance under the cyclic subgroup of S_{n+1}	Audoux.M5
Whole text	Whole text	Provide specific locations in the text for ambiguous references	Raynor.1.2
Introduction	Introduction	Clarify that the question of what symmetric monoidal categories produce how many Vassiliev invariants is open, and will not be resolved in Chapter 3	Raynor.1.2 (final point)
Section 1.2 pictoral representation of strata	Section 1.2 pictoral representation of strata	Label some chambers of the stratum with (singular) knots, give an example of the sign of a singularity. In combination with the picture in the introduction, this should suffice to be clear. Figure now appears on the same page as the worked example, one small paragraph away so I believe a Figure label in not necessary and gets in the way of a consistent style, especially as this would then be the only labelled figure in the text.	Raynor.1.2
Definition 1.2.5 and surrounding text	Definition 1.2.5 and surrounding text	Rephrase Definitions 1.2.5, add the word closed in multiple places to specify that the singular isotopies are closed	Raynor.1.2
Definition 1.2.9	Definition 1.2.9	Make explicit that (a) and (b) define the same objects. For (a) indeed the reviewer is correct to note that there is an equivalent result present, but I believe it is important to include this in the definition environment, as in the literature (specifically frequently in works of Bar-Natan) this is how chord diagrams are defined and the equivalence is left implicit. For (b), we are defining the function σ so this is a valid part of the definition. Example 1.2.10 has been labelled to indicate which double points correspond to which chords.	Raynor.1.2

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Table 1: List of revisions made based on corrections, suggestions or comments in review reports (Continued)

Section 1.2	Section 1.2	Include more explicit references to the Questions and Answer environments of Section 2.1, including referencing in the rest of the chapter.	Raynor.1.2
Proof of Proposition 1.3.4 Proof of Proposition 1.3.5 Proof of Lemma 1.3.10 Remark 1.3.13	Proof of Proposition 1.3.4 Proof of Proposition 1.3.5 Proof of Lemma 1.3.10 Remark 1.3.13	State explicitly assumption that k_1 and k_2 (etc.) are genuine knots (not linear combinations), and explicitly state that results still hold by linearity	Raynor.1.3
-	Definition 2.1.4 Definition 2.1.5	Add a Definition for the **vector space** of Jacobi diagrams and a Proposition and proof sketch that they are isomorphic to the vector space of chord diagrams	Raynor.2.1
	Before Definition 2.1.4	Move the sentence referencing Jacobi diagrams to after their definition	Raynor.2.1
-	Definition 2.1.9	Add definition of linear Jacobi diagram before (old numbering) Proposition 2.1.7	Raynor.2.1
Proof of Proposition 2.1.7	Proof of Proposition 2.1.10	Rephrase in terms of "connected components of the original linear diagram" for clarity	Raynor.2.1
After Examples 2.3.4	After Examples 2.3.5	Add inclination and reference for abelian categories definition	Raynor.2.3
Theorem 2.3.6	Theorem 2.3.7	Add reminder that k denotes the monoidal unit of $\mathcal{C}$	Raynor.2.3
Definition 3.2.1 Paragraph after Definition 3.2.2	Definition 3.2.1 Paragraph after Definition 3.2.2 Raynor.3.2	Define how to count the m in m-singular long knots (i.e. the filtration). Add comment that the relation between double points and {positive, negative} semi-virtuals is homogeneous.	
Definition 3.2.5	Definition 3.2.6	Make clearer what is meant by any AS/IHX/STU relation. (For STU we now explicitly provide all three. This is also somewhat resolved as "any" carries less options now that all vertices are two-in-one-out which came from other feedback.)	Raynor.3.2
Definition 3.4.1 Definition 3.4.2	Definition 3.4.1 Definition 3.4.2	Clarify that the object monoid rather than the prop is generated by a set of colours	Raynor.3.2
Proof of 3.4.4	Proof of 3.4.4	The suggestion was to remind the reader of the two-in-one-out rule. However in implementing the suggestions of Audoux, this has dissapears and only such vertices are allowed by definition. So instead we provide at the start of the proof a paragraph explaining how in the proof, whenever terms with non-matching composition signatures (domain/codomain) are encountered they dissapear, and why this follows from linearity of $\oplus$ and the definition of y.	Raynor.3.2

Table 2: List of issues, comments or questions raised by the reviewers which are acknowledged but no modifications resulted

Relevant point(s) in report(s)	**Explanation**
Auduou.m4: why introduce the notation Q' when it's soon shown to be equal to $P = \partial Q$?	For the same reason we use the notation f and F when learning/proving the fundamental theorem of calculus: not to confuse the notation by a choice which would suggest the hypothesis. Also, to be consistent with the literature, specifically Willterton, which is cited.
Auduox.m5: Remark 1.2.18 is intriguing but uinnformative	Indeed, it is an open question to explain this, so there is nothing informative to say. However I feel that this intriguing point deserves to be stated.
Audoux.m11: why use the words covariant and contravariant?	They are standard enough, and could be useful for readers of less categorical backgrounds. Furthermore they are used a few more times in similar ways in the thesis.
Audoux.m29: "does it means that the universal welded weight system constructed in Section 3.4 is in some sense contained in $\mathfrak{gl}_{1\|1}$ in some way?"	Yes, in some way. This is mentioned in the final paragraph (and additionally a reference is now included).
Audoux.M4: concatenation of welded long knots as an example of definitions appearing late or not at all.	I accept and appreciate this point in general (all other examples listed, and some not listed have led to changes), but I think this is acceptable — for example some basic knot theory notions are also missing in Chapter 1, and the audience is assumed to know some basic topology. Other than this specific instance, the rest of the suggestions of Audoux.M4 were implemented.
Audoux.m16: "I am intrigued by assumption (e)..." (the S_n action in the definition of an operad).	I still believe this to be correct. There is already an S_n action generated by τ from the symmetric monoidal category $\mathcal{C}$. Futhermore, the other examiner who is an expert in this topic had no qualms with this definition.
Raynor.typo2: "...are used but not identified."	These are consistent with the notation for the rest of Chapter.